

Verilog HDL

HDL gives output as logic circuit

Assignments

Assignments are used to assign values into nets and variables.

Three types:

* Continuous, procedural, procedural continuous

Continuous assignment

> Assign a value to net (wire) continuously, whenever the input to the assignment statement change

Ex: assign $c = a + b$

> It only used in combinational logic circuit

RTL components

> Combinational logic

> Sequential logic

> Registers

> Data + control path

> FSMs

> Timing: setup/hold, clocking

> Reset strategies (sync/async)

Digital logic fundamentals

- > Logic gates are the fundamental building blocks of digital electronics.
- > They take one or more binary inputs (0 and 1) to produce single binary output.

Basic logic gates

- AND Gate $\rightarrow A \cdot B = Y$
Output 1 if both inputs are 1.
- OR Gate $\rightarrow A + B = Y$
Output 1 if ^{at least} one input is 1.
- NOT Gate $\rightarrow Y = \bar{A}$
It takes single input and reverses it.

Universal gate

- NAND Gate $\rightarrow Y = \overline{A \cdot B}$
This is an AND gate followed by NOT gate.
- NOR Gate $\rightarrow Y = \overline{A + B}$
This is an OR gate followed by NOT gate.

The special gate

- XOR gate $\rightarrow Y = A \oplus B$
Output 1 if the inputs are different.

Simulation results

By observing the simulation results, when the simulation waveform is formed, It shows clock pulse wave as 0 and 1 uniformly. and reset time have also ~~be~~ formed.